

15. Discuss in what sense the equivalences of Theorem 2.13 (page 117) form the basis of an algorithm which, given ϕ , pushes quantifiers to the top of the formula's parse tree. If the result is ψ , what can you say about commonalities and differences between ϕ and ψ ?
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Exercises 2.4

- * 1. Consider the formula $\phi \stackrel{\text{def}}{=} \forall x \forall y Q(g(x, y), g(y, y), z)$, where Q and g have arity 3 and 2, respectively. Find two models $\mathcal{M}$ and $\mathcal{M}'$ with respective environments l and l' such that $\mathcal{M} \models_l \phi$ but $\mathcal{M}' \not\models_{l'} \phi$.
2. Consider the sentence $\phi \stackrel{\text{def}}{=} \forall x \exists y \exists z (P(x, y) \wedge P(z, y) \wedge (P(x, z) \rightarrow P(z, x)))$. Which of the following models satisfies ϕ ?
- (a) The model $\mathcal{M}$ consists of the set of natural numbers with $P^{\mathcal{M}} \stackrel{\text{def}}{=} \{(m, n) \mid m < n\}$.
 - (b) The model $\mathcal{M}'$ consists of the set of natural numbers with $P^{\mathcal{M}'} \stackrel{\text{def}}{=} \{(m, 2 * m) \mid m \text{ natural number}\}$.
 - (c) The model $\mathcal{M}''$ consists of the set of natural numbers with $P^{\mathcal{M}''} \stackrel{\text{def}}{=} \{(m, n) \mid m < n + 1\}$.
3. Let P be a predicate with two arguments. Find a model which satisfies the sentence $\forall x \neg P(x, x)$; also find one which doesn't.
4. Consider the sentence $\forall x (\exists y P(x, y) \wedge (\exists z P(z, x) \rightarrow \forall y P(x, y)))$. Please simulate the evaluation of this sentence in a model and look-up table of your choice, focusing on how the initial look-up table l grows and shrinks like a stack when you evaluate its subformulas according to the definition of the satisfaction relation.
5. Let ϕ be the sentence $\forall x \forall y \exists z (R(x, y) \rightarrow R(y, z))$, where R is a predicate symbol of two arguments.
- * (a) Let $A \stackrel{\text{def}}{=} \{a, b, c, d\}$ and $R^{\mathcal{M}} \stackrel{\text{def}}{=} \{(b, c), (b, b), (b, a)\}$. Do we have $\mathcal{M} \models \phi$? Justify your answer, whatever it is.
 - * (b) Let $A' \stackrel{\text{def}}{=} \{a, b, c\}$ and $R^{\mathcal{M}'} \stackrel{\text{def}}{=} \{(b, c), (a, b), (c, b)\}$. Do we have $\mathcal{M}' \models \phi$? Justify your answer, whatever it is.
- * 6. Consider the three sentences

$$\begin{aligned}\phi_1 &\stackrel{\text{def}}{=} \forall x P(x, x) \\ \phi_2 &\stackrel{\text{def}}{=} \forall x \forall y (P(x, y) \rightarrow P(y, x)) \\ \phi_3 &\stackrel{\text{def}}{=} \forall x \forall y \forall z ((P(x, y) \wedge P(y, z) \rightarrow P(x, z)))\end{aligned}$$

which express that the binary predicate P is reflexive, symmetric and transitive, respectively. Show that none of these sentences is semantically entailed by the other ones by choosing for each pair of sentences above a model which satisfies these two, but not the third sentence – essentially, you are asked to find three binary relations, each satisfying just two of these properties.

7. Show the semantic entailment $\forall x \neg \phi \models \neg \exists x \phi$; for that you have to take any model which satisfies $\forall x \neg \phi$ and you have to reason why this model must also satisfy $\neg \exists x \phi$. You should do this in a similar way to the examples in Section 2.4.2.
- * 8. Show the semantic entailment $\forall x P(x) \vee \forall x Q(x) \models \forall x (P(x) \vee Q(x))$.
9. Let ϕ and ψ and η be sentences of predicate logic.
 - (a) If ψ is semantically entailed by ϕ , is it necessarily the case that ψ is not semantically entailed by $\neg \phi$?
 - * (b) If ψ is semantically entailed by $\phi \wedge \eta$, is it necessarily the case that ψ is semantically entailed by ϕ and semantically entailed by η ?
 - (c) If ψ is semantically entailed by ϕ or by η , is it necessarily the case that ψ is semantically entailed by $\phi \vee \eta$?
 - (d) Explain why ψ is semantically entailed by ϕ iff $\phi \rightarrow \psi$ is valid.
10. Is $\forall x (P(x) \vee Q(x)) \models \forall x P(x) \vee \forall x Q(x)$ a semantic entailment? Justify your answer.
11. For each set of formulas below show that they are consistent:
 - (a) $\forall x \neg S(x, x), \exists x P(x), \forall x \exists y S(x, y), \forall x (P(x) \rightarrow \exists y S(y, x))$
 - * (b) $\forall x \neg S(x, x), \forall x \exists y S(x, y),$
 $\forall x \forall y \forall z ((S(x, y) \wedge S(y, z)) \rightarrow S(x, z))$
 - (c) $(\forall x (P(x) \vee Q(x))) \rightarrow \exists y R(y), \forall x (R(x) \rightarrow Q(x)), \exists y (\neg Q(y) \wedge P(y))$
 - * (d) $\exists x S(x, x), \forall x \forall y (S(x, y) \rightarrow (x = y))$.
12. For each of the formulas of predicate logic below, either find a model which does not satisfy it, or prove it is valid:
 - (a) $(\forall x \forall y (S(x, y) \rightarrow S(y, x))) \rightarrow (\forall x \neg S(x, x))$
 - * (b) $\exists y ((\forall x P(x)) \rightarrow P(y))$
 - (c) $(\forall x (P(x) \rightarrow \exists y Q(y))) \rightarrow (\forall x \exists y (P(x) \rightarrow Q(y)))$
 - (d) $(\forall x \exists y (P(x) \rightarrow Q(y))) \rightarrow (\forall x (P(x) \rightarrow \exists y Q(y)))$
 - (e) $\forall x \forall y (S(x, y) \rightarrow (\exists z (S(x, z) \wedge S(z, y))))$
 - (f) $(\forall x \forall y (S(x, y) \rightarrow (x = y))) \rightarrow (\forall z \neg S(z, z))$
 - * (g) $(\forall x \exists y (S(x, y) \wedge ((S(x, y) \wedge S(y, x)) \rightarrow (x = y)))) \rightarrow$
 $(\neg \exists z \forall w (S(z, w)))$.
 - (h) $\forall x \forall y ((P(x) \rightarrow P(y)) \wedge (P(y) \rightarrow P(x)))$
 - (i) $(\forall x ((P(x) \rightarrow Q(x)) \wedge (Q(x) \rightarrow P(x)))) \rightarrow ((\forall x P(x)) \rightarrow (\forall x Q(x)))$
 - (j) $((\forall x P(x)) \rightarrow (\forall x Q(x))) \rightarrow (\forall x ((P(x) \rightarrow Q(x)) \wedge (Q(x) \rightarrow P(x))))$
 - (k) Difficult: $(\forall x \exists y (P(x) \rightarrow Q(y))) \rightarrow (\exists y \forall x (P(x) \rightarrow Q(y)))$.

Exercises 2.5

1. Assuming that our proof calculus for predicate logic is sound (see exercise 3 below), show that the validity of the following sequents cannot be proved by finding for each sequent a model such that all formulas to the left of $\vdash$ evaluate to **T** and the sole formula to the right of $\vdash$ evaluates to **F** (explain why this guarantees the non-existence of a proof):